

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		- Proton or delta plus – accept symbols - Leptons - Antiquarks - Weak	4			4		
	(b)		Charge shown to be 0 (i.e. $-\frac{2}{3} + \frac{1}{3} + \frac{1}{3}$) seen (1) Antineutron / antidelta zero (1) accept symbols Don't accept antineutrino	2			2		
	(c)	(i)	Baryon: $1 + 1 = 1 + (-1) + 1 + 0$ Not conserved (1) Lepton: $0 + 0 = 0 + 0 + 0 + 1$ Not conserved (1) Charge: $1 + 0 = 1 + (-1) + 1 + 0$ Conserved (1)		3		3		
		(ii)	Baryon number of pion = 0, so <u>baryon number still not conserved</u> , [removing neutrino will balance lepton number]			1	1		
			Question 7 total	6	3	1	10	0	0